

Pierre Arbaji

Engineering Portfolio

Mechanical design • Structures • Composites & manufacturing

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McGill University, expected April 2027

pierre.arbaji@mail.mcgill.ca • 438-835-1116

LinkedIn

I design, analyse, build and test load-bearing structures. Most of that work has been on rocket hardware (composite airframes, machined metallic interfaces, a pressure vessel), because student rocketry hands the entire design loop to undergraduates: requirements, sizing, finite element analysis, tooling, manufacturing, testing and flight. Internships in aerostructures at Airbus and in advanced composite structures at Bombardier, and undergraduate research on polymer actuators for architected metamaterials, have extended the same interest in how structures and materials behave and how far they can be pushed.

This portfolio documents the engineering behind that work rather than repeating my CV: the requirements each design answered, the decisions and trade-offs, the analysis and how it was checked, and what testing and flight showed.

Project index

	Project and context	Methods	Key evidence	p.
1	Project Luminor aerostructures Hybrid rocket, McGill Rocket Team; lead of a ~50-student subteam	Architecture, composite design, tooling and prepreg layup, FEA, CNC interfaces, integration	3rd place, Launch Canada (Advanced); structure recovered undamaged; 13.35 kg vs. 18.1 kg budget	2
2	Classical laminate theory design tool Individual project, used to size Luminor's composites	CLT (<i>ABD</i>), six failure criteria, shell and panel buckling, independent verification	24 benchmark quantities within 0.05% of an independent Python code; sized laminates flew without damage	7
3	Structural redesign and optimization Two flight components for a higher-apogee vehicle	Load-path redesign, topology optimization, design for 3-axis machining, mesh convergence	Recovery assembly 85% lighter; cage bending FoS 1.50 → 2.10 with 17.7% less mass	10
4	Research: supercoiled polymer muscles Architected Metamaterials Group; undergraduate RA	Fabrication, electro-thermal testing, failure analysis, integration design	12% stroke against a 2.84 N load (2.0 N required); co-author	15
5	Aircraft conceptual design AERO 520 long-range business jet; team of four	Constraint sizing, performance, stability, structural layout	Field lengths 5,234 / 5,206 ft vs. 6,000 ft limit; 1.9% range shortfall traced to cruise L/D	18

Current work (2026–27): Design and Optimization Technical Director, McGill Rocket Team; capstone project on the design, manufacture and ground testing of a one-piece composite fin can for the team's supersonic two-stage vehicle.

Software used in this portfolio: Siemens NX, SolidWorks, CATIA (CAD); Nastran, Ansys (FEA); MATLAB, Python, Excel; OpenRocket, RocketPy; XFOIL, AAA (Advanced Aircraft Analysis).

1 Project Luminor: Rocket Aerostructures

McGill Rocket Team • Aerostructures Subteam Lead, May 2025 – Aug 2026 • Subteam of ~50 students

Vehicle. Single-stage hybrid rocket, significantly larger than the team’s previous vehicle, built as forward, propulsion and aft assemblies.

My role. I led the aerostructures subteam from architecture through manufacturing, integration and flight, coordinating structural decisions with propulsion, avionics, recovery, payload and launch operations.

Outcome. Luminor placed 3rd in the Advanced category at Launch Canada, and the structure was recovered with no damage. Aerostructures mass was 13.35 kg against an 18.1 kg budget (26% under), and the full vehicle was repeatedly assembled in under one hour.

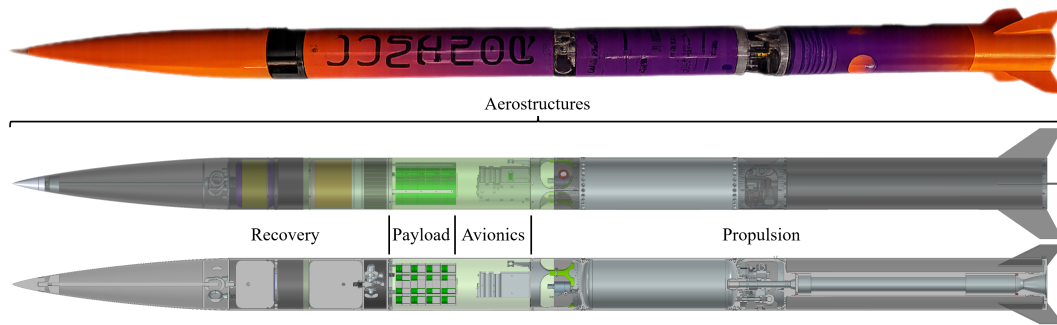


Figure 1: Project Luminor: final vehicle, CAD architecture and structural layout.

Problem and requirements

The previous competition showed that field assembly was a weak point, and Luminor’s larger airframe raised both loads and mass. The structural subsystem therefore had to satisfy four coupled requirements:

1. Carry flight, propulsion and recovery loads with a minimum factor of safety of 1.5.
2. Stay within an 18.1 kg aerostructures mass budget.
3. Keep the vehicle’s static stability margin within the lower and upper limits across the expected wind range.
4. Allow the vehicle to be manufactured, transported and assembled as independent modules.

Architecture and key design decisions

The external airframe is composite; internal structures and all primary joints are machined aluminium, with additively manufactured parts limited to non-primary structure. The table records the decisions that shaped the subsystem and the evidence behind each.

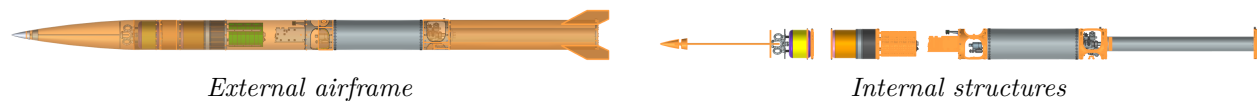


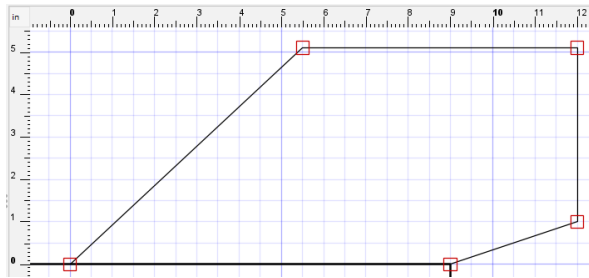
Figure 2: Aerostructures architecture: CFRP/GFRP airframe, honeycomb fins, RADAX couplers, payload and propulsion interfaces.

Decision	Rationale	Evidence
CFRP for most of the airframe; GFRP around avionics	Stiffness and mass efficiency where they govern; radio transparency where antennas sit	Laminates sized with the CLT tool (Project 2); flew without damage
CFRP / aramid-honeycomb sandwich fins, swept clipped delta	High bending stiffness at low mass; planform iterated against static-stability and flutter constraints	Stability margin ≥ 1.5 calibers in winds up to ~ 17 mph
Machined Al 6061-T6 RADAX couplers replacing the tube-and-leg joint	Higher joint stiffness, fewer structural parts, faster assembly	FEA: bending FoS 1.50, ejection FoS 2.82
Modular forward / propulsion / aft assemblies	Independent manufacture and transport; shorter field assembly	Assembly under 1 h in repeated rehearsals
Drogue deployment triggered before apogee	Lesson from the previous vehicle; gives the parachute time to inflate	Dynamic model $\rightarrow$ 30.6 m/s velocity trigger

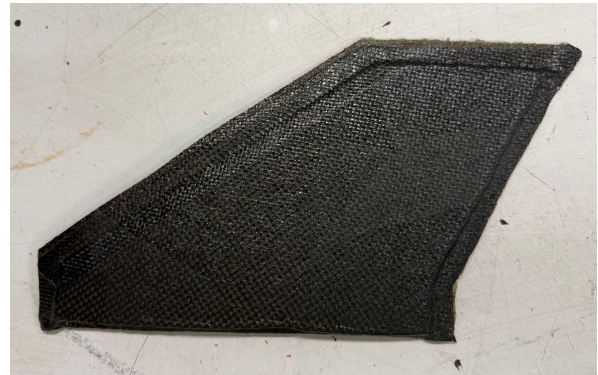
Composite design and manufacturing

The fins use three CFRP plies on each face of an aramid honeycomb core. Aerodynamic loads enter the airframe through a through-the-wall attachment, structural epoxy and continuous tip-to-tip reinforcement. Airframe tube laminates were sized with the classical laminate theory tool described in Project 2.

The team produced its own tooling. For the nose cone I worked through the complete chain: CAD pattern $\rightarrow$ large-format PLA print $\rightarrow$ glass-fibre wet-layup female mould (sealed and release-treated) $\rightarrow$ CFRP prepreg layup $\rightarrow$ vacuum consolidation $\rightarrow$ oven cure. Airframe tubes were laid up on reusable aluminium mandrels and consolidated with shrink tape; flat laminates were vacuum-bagged and machined into internal parts, extending composite construction beyond the primary airframe.



Final fin geometry



Cured honeycomb-sandwich test fin

Figure 3: Composite fin design and manufacture.

Structural analysis

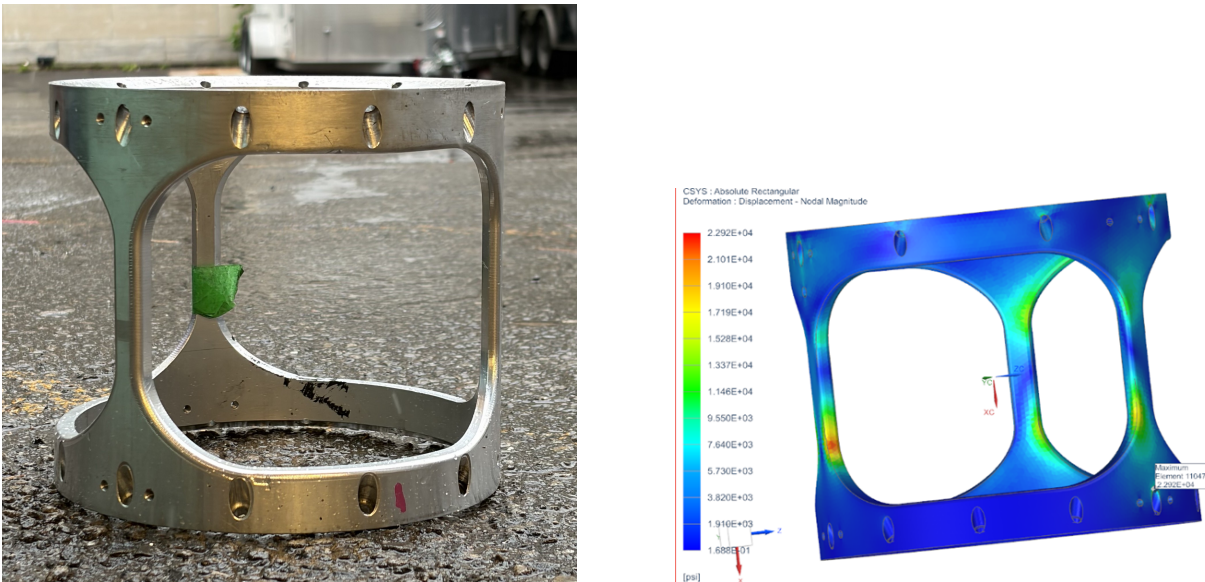
Each primary component was sized by hand calculation and checked by FEA under its governing cases: axial thrust and acceleration, flight bending, aerodynamic fin loads, pneumatic and pyrotechnic recovery loads, internal pressure, and local fastener loads.



3D-printed pattern *Glass-fibre female mould* *CFRP tube prepreg layup*
Figure 4: In-house tooling and layup: nose-cone pattern and mould, airframe tube on its mandrel.

Component	Governing load case	FoS	Verification
RADAX coupler	Vehicle bending	1.50	FEA
RADAX coupler	Pneumatic ejection	2.82	FEA
Payload plate	12 kN pneumatic ejection (122 MPa peak)	1.97	FEA, mesh-converged
Engine retainer	Motor thrust (136.5 MPa peak)	1.77	FEA
Oxidizer tank	Internal pressure, MEOP 6.21 MPa	2.22	ASME sizing, FEA, hydrostatic proof test

Factors of safety on yield; Al 6061-T6 taken as 241 MPa (35 ksi) throughout this portfolio. Minimum required: 1.5.



Machined RADAX coupler *Bending FEA (elemental stress)*
Figure 5: RADAX coupler: governing bending case, FoS 1.50; pneumatic ejection case, FoS 2.82.

The payload plate shows the verification procedure used on the primary interfaces: stress was re-evaluated as element size was progressively reduced until the maximum converged at approximately 122 MPa, giving a factor of safety of 1.97 under the 12 kN ejection load.

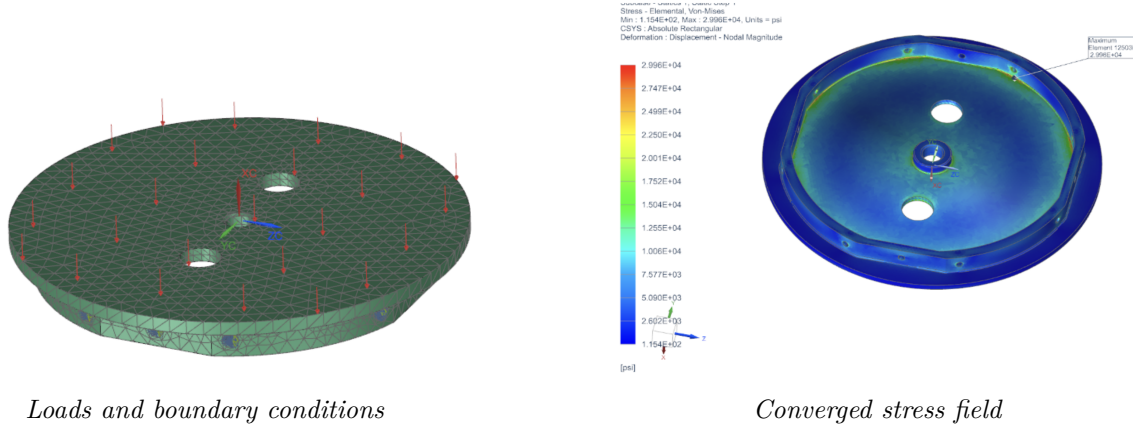


Figure 6: Payload plate under the 12 kN pneumatic ejection load: 122 MPa maximum, FoS 1.97.

Oxidizer tank integration

The oxidizer tank is part of the vehicle's primary structure, so its design had to satisfy pressure-vessel and airframe requirements together. It uses a machined Al 6061-T6 cylinder with Al 7075-T6 torispherical heads, sized to ASME pressure-vessel rules; FEA of the heads and pipe geometry located stress concentrations and identified where mass could be removed. The final tank has a 20.62 cm outer diameter, a 6.21 MPa maximum expected operating pressure (MEOP), 13 kg oxidizer capacity and an overall factor of safety of 2.22. It was CNC machined, sealed with radial fasteners and O-rings, hydrostatically proof-tested to $1.5 \times \text{MEOP}$ (≈ 9.3 MPa), and then passed four cold-flow and two hot-fire tests.



Figure 7: Manufactured oxidizer tank assembly and machined heads.

Vehicle-level stability and recovery

Structural mass, fin geometry and recovery architecture feed directly into flight behaviour, so they were iterated against vehicle-level models. In OpenRocket, the final configuration kept a minimum predicted stability margin of 1.5 calibers in winds up to about 17 mph while staying below the upper stability limit. For recovery, a dynamic model of the descent, including atmospheric density, drag-area growth during parachute inflation and deployment delay, was used to set a pre-apogee drogue trigger at approximately 30.6 m/s. Landing dispersion was then estimated in RocketPy by sweeping wind speed and direction.

Integration and flight

Parts from five processes (prepreg CFRP/GFRP, composite tooling, honeycomb sandwich, CNC machining and 3D printing) had to meet at shared interfaces. Repeated assembly rehearsals exposed integration problems before competition, and the final assembly was repeatedly completed in under

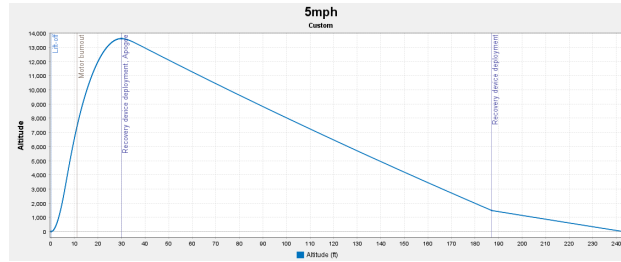


Figure 8: Vehicle-level flight model in OpenRocket.

one hour, confirming the purpose of the modular architecture. At Launch Canada the vehicle flew, placed 3rd in the Advanced category, and was recovered with no structural damage.



Figure 9: Integrated Project Luminor vehicle.

What this project shows. Ownership of a large mechanical system across its full life cycle, and a design method that does not rely on one tool: hand sizing, FEA with convergence checks, physical tests (hydrostatic proof, cold flow, hot fire) and finally flight each checked the others.

2 Classical Laminate Theory Design Tool

McGill Rocket Team • Individual project, built as Aerostructures Subteam Lead • Used to size Project Luminor's composite tubes and structures

Composite airframes, fins, nose cones and couplers have no single modulus or strength: each ply responds according to its orientation, and stress has to be tracked ply by ply, in each ply's material axes, under every load case. To decide how thick Luminor's tubes and composite structures needed to be, I built a reusable Excel workbook that automates this chain for preliminary sizing of rocket composite structures, from material properties to a pass/fail verdict, replacing ad-hoc hand calculations and single-purpose spreadsheets with one traceable and independently verified tool.

Scope: a material database (unidirectional and woven carbon, glass and aramid); full CLT including the extensional, coupling and bending stiffness matrices; conversion of flight loads (axial, bending, shear, torque, pressure) into laminate line loads; ply-by-ply strain and stress with six failure criteria; closed-form shell and panel buckling; and verification of every equation against an independent implementation.

Architecture

Thirteen linked sheets form a one-directional pipeline, so any result on the summary dashboard traces back to a single input: materials → laminate builder (ABD) → geometry and loads → CLT solver (mid-plane strain and curvature) → ply stresses → failure and buckling checks → margin of safety. Material data can be edited without touching any downstream formula; named ranges, data validation and conditional formatting let future team members use the tool correctly without reading its internals.

CONFIGURATION		DESIGN LINE LOADS (ultimate)	
Laminate	AF-4in-QI-8ply	Nx (N/m)	-13779.1
Stacking sequence	[0/45/-45/90]s quasi-isotropic carbon airframe wall	Ny (N/m)	0.0
Number of plies	8	Nxy (N/m)	1849.1
Laminate thickness (mm)	1.000	Mx (N.m/m)	0.0
Symmetry	YES - B is negligible	My (N.m/m)	0.0
Balance	YES - no extension/shear coupling	Mxy (N.m/m)	0.0
Component	Airframe tube	eps0_x (microstrain)	-268
Mean radius (mm)	102.63	eps0_y (microstrain)	83
R / t	102.6	gamma0_xy (microstrain)	94
Load case	Max Q - boost + gust	kappa_x (1/m)	0.000
Design factor of safety	1.50		
STRENGTH		STABILITY AND MASS	
Max [sigma_1] (MPa)	36.11	Governing buckling mode	Cylinder - torsion / shear
Max [sigma_2] (MPa)	2.07	Buckling MS	13.46
Max [tau_12] (MPa)	1.58	Design limited by	BUCKLING
Critical ply	8	Equivalent Ex (GPa)	51.51
Critical surface	Bottom	Equivalent Gxy (GPa)	19.53
Governing criterion	Tsai-Wu	Largest deflection (mm)	0.500
Reserve factor RF	36.29	Component mass (kg)	1.0286
Margin of safety MS	34.29	Total structural mass (kg)	2.7819
VERDICT			
Overall margin of safety	13.46		
Status	PASS		

Figure 10: Design summary dashboard: configuration, applied loads, governing failure mode, buckling margin and structural mass for the active laminate.

Theory and independent verification

For an orthotropic ply the reduced stiffnesses are

$$Q_{11} = \frac{E_1}{\Delta}, \quad Q_{12} = \frac{\nu_{12}E_2}{\Delta}, \quad Q_{22} = \frac{E_2}{\Delta}, \quad Q_{66} = G_{12}, \quad \Delta = 1 - \nu_{12}\nu_{21}.$$

Each ply is transformed to laminate axes at its angle θ and integrated through the thickness:

$$A_{ij} = \sum_{k=1}^n \bar{Q}_{ij}^{(k)}(z_k - z_{k-1}), \quad B_{ij} = \frac{1}{2} \sum_{k=1}^n \bar{Q}_{ij}^{(k)}(z_k^2 - z_{k-1}^2), \quad D_{ij} = \frac{1}{3} \sum_{k=1}^n \bar{Q}_{ij}^{(k)}(z_k^3 - z_{k-1}^3).$$

Mid-plane strain and curvature follow from $\{\varepsilon^0; \kappa\} = [A \ B \ D]^{-1}\{N; M\}$, and ply stress from $\sigma^{(k)}(z) = \bar{Q}^{(k)}(\varepsilon^0 + z\kappa)$, evaluated at the bottom, middle and top of every ply to capture the jump at each interface.

Verification. Every equation was checked against a Python reference written independently and structured differently: it builds $\bar{Q}$ with the matrix triple product $\bar{Q} = T^{-1}QRTT^{-1}$ rather than the expanded polynomial form used in the workbook. Across 24 benchmark quantities (the full A , B and D matrices, mid-plane strain, ply stress and critical buckling load), agreement was better than 0.05% on every value.

PLY DEFINITION (input)					GEOMETRY (m)	
Ply #	Material	Angle (deg)	t override (mm)	t used (mm)	t (m)	z_bot (m)
1	CF-UD-T700-Epoxy	0.0		0.125	0.000125	-0.000500
2	CF-UD-T700-Epoxy	45.0		0.125	0.000125	-0.000375
3	CF-UD-T700-Epoxy	-45.0		0.125	0.000125	-0.000250
4	CF-UD-T700-Epoxy	90.0		0.125	0.000125	-0.000125
5	CF-UD-T700-Epoxy	90.0		0.125	0.000125	0.000000
6	CF-UD-T700-Epoxy	-45.0		0.125	0.000125	0.000125
7	CF-UD-T700-Epoxy	45.0		0.125	0.000125	0.000250
8	CF-UD-T700-Epoxy	0.0		0.125	0.000125	0.000375

ABD STIFFNESS MATRIX

	eps0_x	eps0_y	gamma0_xy	kappa_x	kappa_y	kappa_xy
Nx (N/m)	5.705E+07	1.778E+07	0.000E+00	-9.095E-13	2.842E-14	0.000E+00
Ny (N/m)	1.778E+07	5.705E+07	1.863E-09	2.842E-14	-6.821E-13	0.000E+00
Nxy (N/m)	0.000E+00	1.863E-09	1.963E+07	0.000E+00	0.000E+00	-2.842E-14
Mx (N.m/m)	-9.095E-13	2.842E-14	0.000E+00	7.974E+00	1.246E+00	4.971E-01
My (N.m/m)	2.842E-14	-6.821E-13	0.000E+00	1.246E+00	2.008E+00	4.971E-01
Mxy (N.m/m)	0.000E+00	0.000E+00	-2.842E-14	4.971E-01	4.971E-01	1.400E+00

Figure 11: Laminate builder: ply stacking table and computed ABD matrix.

Failure and stability

Six criteria are evaluated at every ply surface: maximum stress, maximum strain, Tsai–Hill, Tsai–Wu, Hashin and Puck. The Tsai–Wu index is quadratic and non-homogeneous in stress, so the reserve factor is the positive root of the interaction polynomial:

$$a\lambda^2 + b\lambda - 1 = 0 \Rightarrow \text{RF} = \frac{-b + \sqrt{b^2 + 4a}}{2a},$$

$$a = F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2, \quad b = F_1\sigma_1 + F_2\sigma_2.$$

The common shortcut $\text{RF} = 1/\text{FI}$ is only valid when the linear terms vanish ($X_t = X_c$, $Y_t = Y_c$). Because the failure polynomial is convex and zero at the origin, $1/\text{FI}$ *overestimates* the reserve factor for any ply with positive margin. That is the non-conservative direction, and exactly the plies a designer is trying to sign off.

For cylindrical shells under axial compression, the classical orthotropic critical load is reduced by the NASA SP-8007 empirical knockdown factor for imperfection sensitivity:

$$N_{x,\text{cr}}^{\text{cl}} = \frac{2}{R} \sqrt{D_{11} \left(A_{11} - \frac{A_{12}^2}{A_{22}} \right)}, \quad \phi = \frac{1}{16} \sqrt{\frac{R}{t}}, \quad \gamma = 1 - 0.901 \left(1 - e^{-\phi} \right).$$

The workbook reports whether a design is buckling-limited or strength-limited, because the fixes are opposite: buckling calls for more bending stiffness D_{11} , strength for more fibre in the governing direction.

Failure Analysis

Design criterion for the dashboard **Most conservative**Tsai-Wu interaction coefficient f1 **-0.50**Puck p(+) perp/para **0.30**Puck p(-) perp/para **0.25**

GOVERNING RESULT

Minimum reserve factor RF **35.29**Margin of safety MS = RF - 1 **34.29**Governing ply **8**Governing surface **Bottom**Governing criterion **Tsai-Wu**Verdict **PASS - pos**

PLY				STRESS - ply axes			STRAIN - ply axes		
Ply	Angle	Surface	Act	sigma_1 (MPa)	sigma_2 (MPa)	tau_12 (MPa)	eps_1	eps_2	gamma_12
1	0.0	Bottom	1	-36.11	0.00	0.42	-2.675E-04	8.339E-05	9.418E-05
1	0.0	Mid	1	-36.11	0.00	0.42	-2.675E-04	8.339E-05	9.418E-05
1	0.0	Top	1	-36.11	0.00	0.42	-2.675E-04	8.339E-05	9.418E-05
2	45.0	Bottom	1	-6.48	-1.31	1.58	-4.497E-05	-1.391E-04	3.509E-04
2	45.0	Mid	1	-6.48	-1.31	1.58	-4.497E-05	-1.391E-04	3.509E-04
2	45.0	Top	1	-6.48	-1.31	1.58	-4.497E-05	-1.391E-04	3.509E-04

Ply-by-ply failure index and margin

Figure 12: Failure and buckling sheets: governing ply, criterion and margin for the active load case.

STABILITY CHECKS					
Buckling mode	Critical load	Unit	Applied	Ratio (crit/applied)	MS = ratio - 1
Cylinder - axial compression	228030.4	N/mm	13779.1	16.55	15.55 YES
Cylinder - bending compression	256525.9	N/mm	6800.3	38.16	37.16 no
Cylinder - torsion / shear	26738.5	N/mm	1849.1	14.46	13.46 YES
Cylinder - external pressure	10557.2	Pa	0.0		
Flat panel - uniaxial compression (fins, stiffened skins)		N/mm	13779.1		
Flat panel - shear		N/mm	1849.1		
Global Euler column	1672256.5	N	4500.0	371.61	370.61 no

BUCKLING SUMMARY	
Minimum buckling MS	13.46
Governing buckling mode	Cylinder - torsion / shear
Minimum strength MS (Failure sheet)	34.29
Design is limited by	BUCKLING
Overall structural MS	13.46
Verdict	PASS

Shell and panel buckling checks

Application and flight

The tool set the laminate thicknesses of Luminor's composite airframe tubes and structures. Luminor flew at Launch Canada and its composite structure was recovered with no damage. That confirms the sized laminates at flight loads on a pass/fail basis; a quantitative validation would compare coupon or tube tests with the predicted margins, which is the natural next step for the tool.

What this project shows. Graduate-level composite mechanics implemented end to end, and the discipline to verify a tool before trusting it with flight hardware: an independent, differently structured implementation; correct reserve factors for interactive criteria; and buckling knockdowns treated as visible, first-class inputs rather than buried constants.

3 Structural Redesign: Load-Path and Topology Optimization

McGill Rocket Team • Aerostructures Subteam Lead • Redesign for the team's next, higher-apogee vehicle

Two flight-critical parts were redesigned with two different strategies. For the recovery interface, the load path itself was changed so that material which carried no load could be removed. For the propulsion cage, the architecture was kept and topology optimization was used to redistribute material toward the dominant bending load, followed by a manual redesign for three-axis machining. Both designs had to keep their existing interfaces to the propulsion and recovery hardware and meet the team's minimum factor of safety of 1.5.

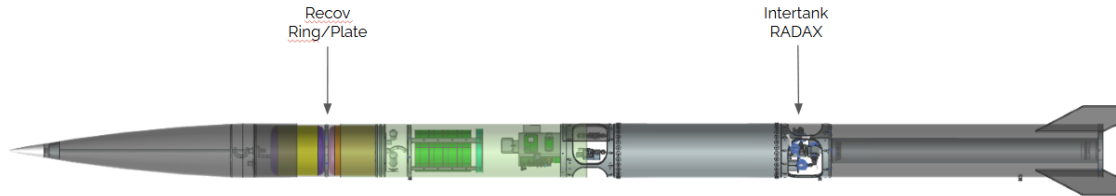
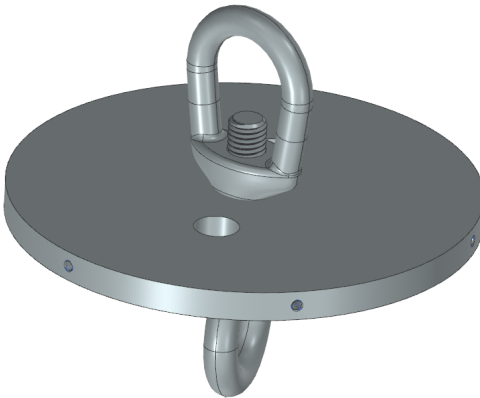


Figure 13: Location of the two redesigned structures within the vehicle.

3.1 Recovery interface: removing material that carried no load

The original interface was a central plate with two heavy eyebolts, one each for the main and drogue parachutes; the assembly weighed 3.4713 lbm (1.575 kg). The centre of the plate lay off the primary load path. The redesign carries the harness loads directly into a machined aluminium ring with two integral attachment regions and an open centre; two carbon-fibre plates bonded to opposite faces retain the parachute assembly so it cannot pass through the opening.

The ring weighs 0.4075 lbm and the two plates 0.1 lbm together, so the complete interface weighs 0.5075 lbm (0.230 kg): 85.4% lighter than the assembly it replaced.



Original: plate and two eyebolts, 1.575 kg



Redesign: ring and CFRP plates, 0.230 kg

Figure 14: Original and redesigned recovery interfaces.

Verification

The ring was analysed under two recovery load cases with three progressively refined meshes and local mesh controls:

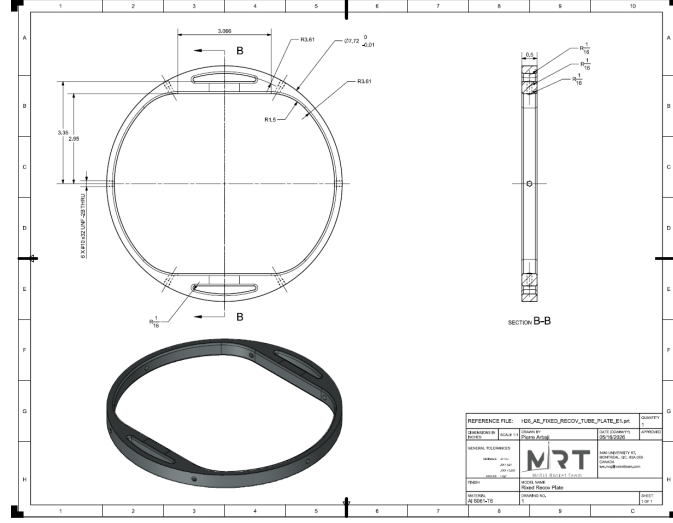


Figure 15: Manufacturing drawing used for CNC fabrication of the recovery ring.

Table 1: Recovery-ring mesh convergence.

Load (lbf)	Global mesh (in)	Local control (in)	Elemental stress (psi)	Nodal stress (psi)
1962	0.20	0.10	17690	25970
1962	0.12	0.06	17510	21780
1962	0.08	0.03	18610	21290
2700	0.20	0.10	3631	4663
2700	0.12	0.06	3742	4417
2700	0.08	0.03	3882	4346

For the governing 1962 lbf (8.73 kN) case, nodal stress changed by 2.3% between the two finest meshes and elemental stress by 6.3%. At the finest mesh the two still differ by about 14%, indicating a local gradient at the peak, so the margin is based on the higher nodal value:

$$FoS_y = \frac{35\,000}{21\,290} = 1.64 \geq 1.5.$$

3.2 Propulsion cage: topology optimization under a manufacturing constraint

The cage sits between the thrust chamber and the oxidizer tank. It carries the primary thrust and bending loads while leaving access to the propulsion valves. The original Al 6061-T6 thin-lattice cage met the current requirements, but its bending factor of safety was 1.50, exactly the team minimum. The next vehicle flies to a higher apogee, which raises the bending load, so bending became the design driver, with mass and manufacturability as secondary objectives. The other three cases already had factors of safety of 4.0 or more.

Table 2: Load cases and margins of the original cage.

Load case	Load (N)	FoS	Governing condition
Thrust	6204	5.61	Launch / maximum acceleration
Bending	2211	1.50	Maximum normal aerodynamic load
Buckling	6204	35.4	Launch / maximum acceleration
Tension	8558	4.00	Worst-case parachute shock

Topology optimization as a load-path guide

The interfaces and load-introduction regions were frozen as non-design space. On the remaining volume, the optimization minimized mass subject to a factor of safety of 2.0 under the bending case, with a mirror-symmetry constraint and a cyclic-symmetry constraint forcing three identical sectors. The result was used as a guide to the load paths, not as a geometry: material was kept around the interfaces and along the dominant paths, and the part was then redrawn by hand to meet manufacturing and integration constraints.

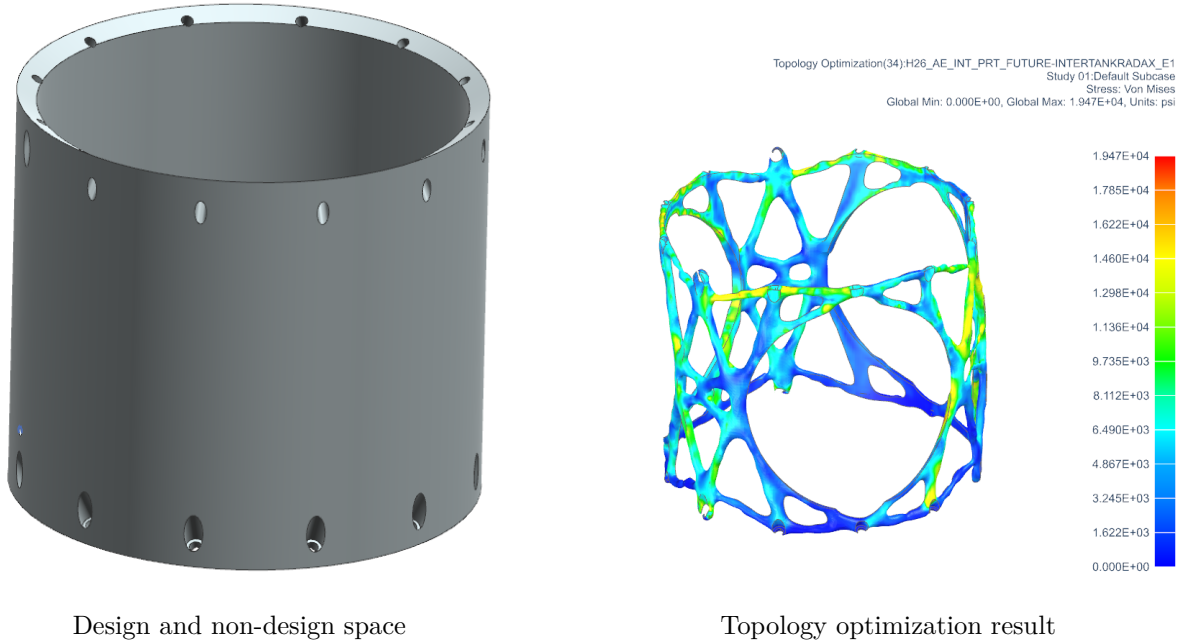


Figure 16: Design space and topology optimization of the structural cage: minimum mass, FoS 2.0 in bending, three identical sectors.

Design for manufacturing

The original lattice required five-axis machining. The redesigned openings were shaped so the principal features can be cut on a conventional three-axis CNC setup while following the load paths from the optimization. Opening sizes were set with the propulsion team: large enough for valve access, no larger.

Verification

Under the governing bending case, the redesigned cage reached a maximum stress of

$$\sigma_{\max} = 1.663 \times 10^4 \text{ psi (114.7 MPa)}, \quad FoS_y = \frac{35\,000}{16\,630} = 2.10,$$

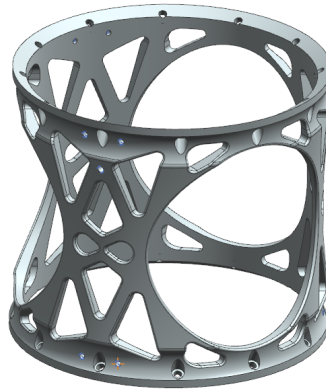
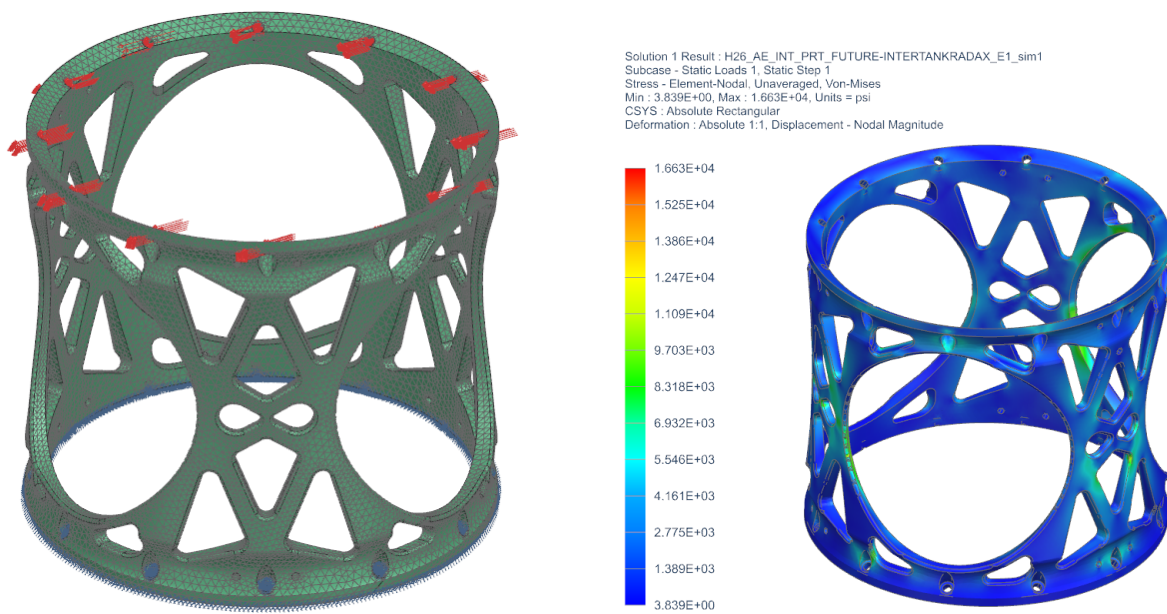


Figure 17: Final cage geometry.

above the 2.0 target. Thrust, buckling and tension were not re-analysed for the new geometry; the redesign addressed bending only.



Loads and constraints

Stress result, redesigned cage

Figure 18: Bending FEA of the redesigned cage: 114.7 MPa maximum, FoS 2.10.

Results

Table 3: Summary of the two redesigns.

Component	Original mass	New mass	Mass change	Factor of safety
Recovery interface	1.575 kg	0.230 kg	−85.4%	1.64
Propulsion cage	0.854 kg	0.703 kg	−17.7%	Bending 1.50 → 2.10

What this project shows. Two routes to a lighter structure: change the load path when material is doing no work, and redistribute material when the architecture is sound but the margin is not. In both cases the optimizer and the FEA informed the design; manufacturing and integration constraints set its final form.

4 Research: Electro-Thermal Supercoiled Polymer Muscles for Bistable Metamaterials

Architected Metamaterials Group, McGill • Undergraduate Research Assistant, Mar 2026 – present • Co-author

Research context. Reprogrammable bistable metamaterials switch between two stable configurations; making them switch on command requires an actuator small enough to embed in the structure. The target here was a laser-cut bistable nylon-foam unit that needs a 2.0 N pull to change state. My role was to fabricate and characterise a supercoiled polymer (SCP) muscle able to supply that force, identify its limits, and design its mechanical integration. I wrote the experimental methodology and results for the project.

A single non-conductive polymer fibre lacked the cross-section to generate 2.0 N reliably, so a dual-fibre plied configuration was selected. Actuation is electro-thermal: a Nichrome 80 resistance wire wrapped around the polymer core provides Joule heating.

Fabrication

1. **Coiling:** two strands of polymer fishing line, each under a 290 g load, twisted until full-length helical self-coiling.
2. **Plying:** the two coiled strands plied about a common axis under a 500 g load.
3. **First anneal:** extended to 110% of relaxed length, 1 h at 160°C.
4. **Second anneal:** after cooling, extended a further 33%, 1 h at 160°C.
5. **Heater:** 34 AWG bare Nichrome 80 wound helically with uniform pitch to avoid shorts and hot spots.

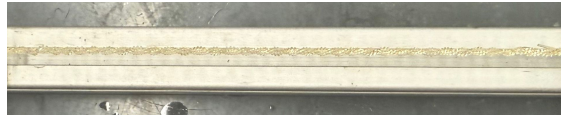


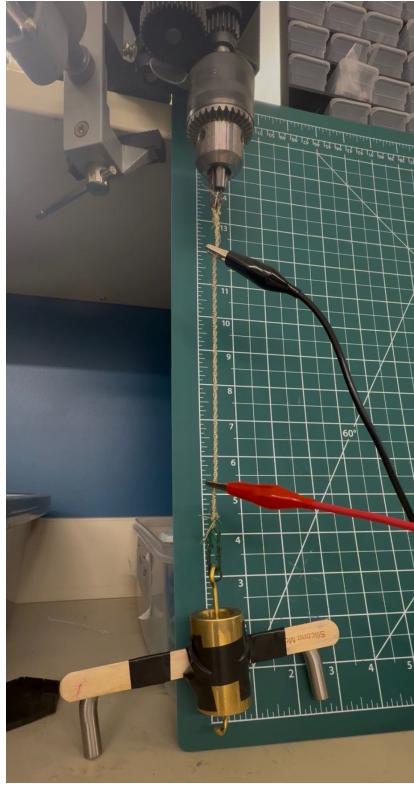
Figure 19: Dual-fibre plied SCP muscle before Nichrome wrapping.

Electro-thermal test

The muscle was suspended vertically under a 290 g dead load with an active heated length of $L_0 = 6.25$ in and driven by a manual voltage sweep from 4.0 V to 8.5 V. It contracted by 0.75 in; at 8.5 V the polymer core ruptured near the centre of the heated segment.

$$\epsilon_a = \frac{\Delta L}{L_0} = \frac{0.75 \text{ in}}{6.25 \text{ in}} = 12.0\%, \quad F_L = M_L g = 0.290 \times 9.81 = 2.84 \text{ N},$$

$$\Delta L = 0.75 \times 0.0254 = 0.01905 \text{ m}, \quad W_{\text{out}} = F_L \Delta L = 2.84 \times 0.01905 = 0.054 \text{ J}.$$



Before current



After current

Figure 20: Muscle under the 290 g test load before and after actuation.**Table 4:** Measured performance.

Quantity	Result
Force during contraction	2.84 N dead load (2.0 N required)
Stroke	19.05 mm (12.0% strain)
Work per stroke	0.054 J
Voltage at rupture	8.5 V

The muscle contracted 12% while lifting 2.84 N, 42% above the force needed to switch the metamaterial, which establishes feasibility for the intended transition.

Failure analysis

Joule heating scales as $P = V^2/R$, so the last steps of the sweep raised heating power sharply. The most likely mechanism is that the core exceeded its thermal limit and, softened under the constant 2.84 N load, yielded and ruptured. Defining a safe operating envelope requires mapping voltage, current, temperature and contraction together, and setting the drive limit below the onset of thermal degradation.

Integration design

The muscle is embedded in the hollow centre of the metamaterial. The main challenge is anchoring a tensioned soft thread at rigid end plates without slip. The proposed termination routes each end

through a 3D-printed end plate and an aluminium double-barrel micro-crimp sleeve, swaged to form a mechanical block that cannot pull through the plate.

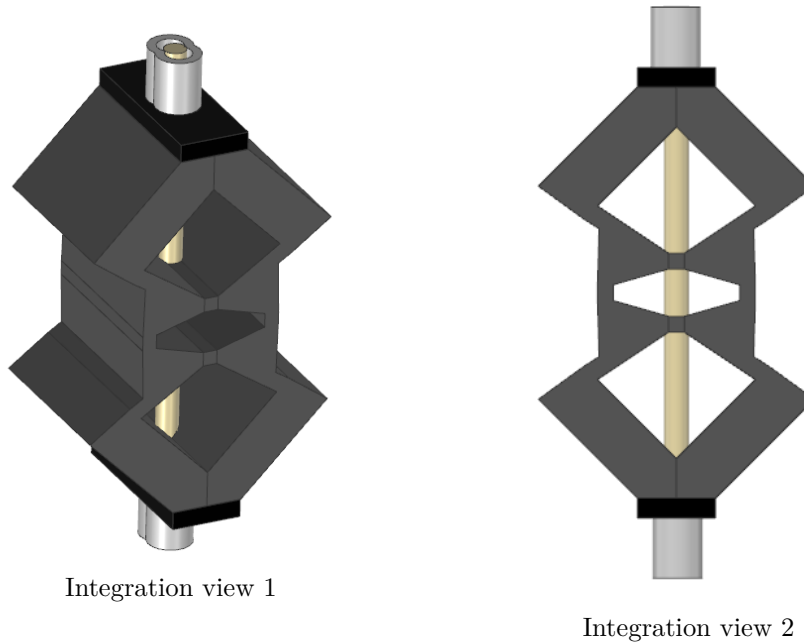


Figure 21: Concept for integrating the SCP muscle into the bistable metamaterial.

Next steps. Current-limit and temperature mapping; repeat tests across several samples; full-assembly tests of the muscle switching the bistable structure.

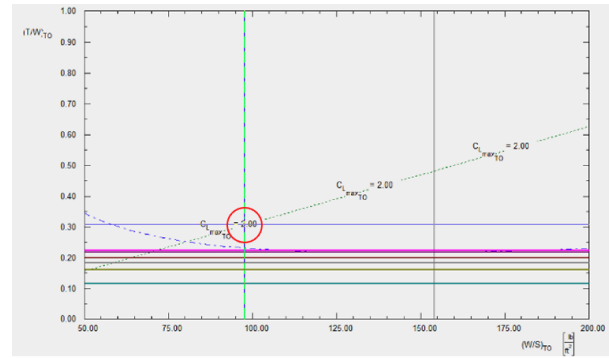
5 Aircraft Conceptual Design: Long-Range Business Jet

McGill AERO 520 (aircraft design) • Team of four • My responsibilities: Led Structural Layout Design

Conceptual and preliminary design of an ultra-long-range business jet for entry into service around 2035, answering a request for proposal (RFP) aimed at the market segment of the Gulfstream G700 and Bombardier Global 7500. Key requirements: 8,000 nmi range with 10% fuel reserves; Mach 0.85 cruise minimum (0.92 target); takeoff and landing field lengths under 6,000 ft (sea level, ISA); FAA Part 25 with flight into known icing; eight passengers, bedroom, shower, galley and three crew.



Final configuration



Matching plot and design point

Figure 22: Configuration and performance-constraint sizing.

Sizing. Takeoff weight was estimated with Roskam's regression method using justified 2035 assumptions ($c_j = 0.40$, $L/D = 14$). In the constraint (matching) plot, stall speed and takeoff distance governed; the design point was placed at their intersection to minimize wing area: $W/S = 97.57 \text{ lb/ft}^2$, $T/W = 0.31$, $S_w = 1,310.25 \text{ ft}^2$.

Configuration. XFOIL/AAA iterations showed an SC(2)-0410 wing could not reach the required cruise $C_L \approx 0.44$; a more cambered SC(2)-0610 root gave $C_L \approx 0.45$. The wing has 35° quarter-chord sweep (critical Mach), taper 0.30 and -5° washout (root-first stall), with a T-tail sized by volume coefficients. Class II weights from three methods (discarding the lowest estimate for conservatism) converged to an MTOW of 95,260 lb and a takeoff thrust requirement of 29,676 lbf, met by two GE Passport engines rubber-scaled by 0.784.

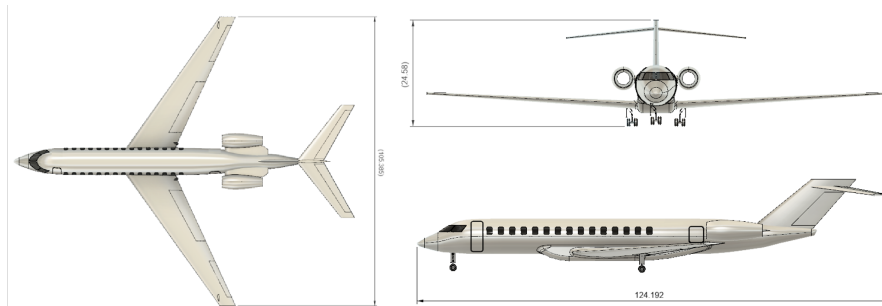


Figure 23: Three-view: 124.2 ft long, 105.53 ft span, 24.58 ft tall.

Structural layout. Fuselage frames at 18 in and longerons at 9 in (tightened to 12 in frames

around the windshield); flat forward and domed aft pressure bulkheads, with the aft bulkhead behind the baggage bay so baggage stays pressurized. The wing box uses spars at 15% and 60% chord, the rear spar placed forward of Roskam's recommended range to make room for flaps and ailerons, with ribs every 24 in, bolted to reinforced fuselage frames.

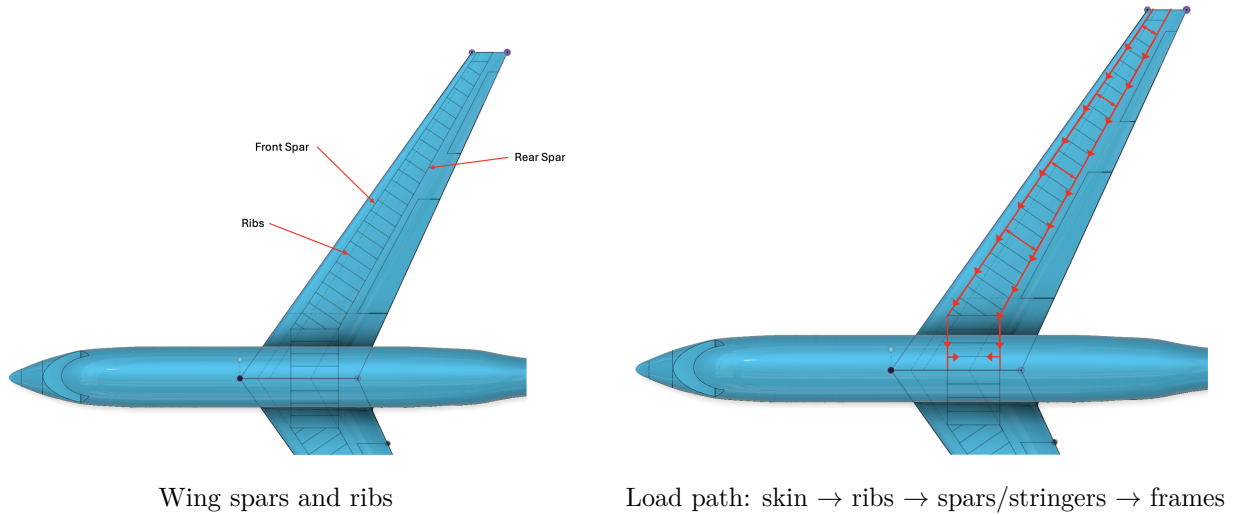


Figure 24: Wing structure and load path.

Performance and stability. Takeoff and landing were integrated segment by segment in MATLAB with velocity-dependent lift, drag and thrust; the balanced field length is the intersection of the accelerate-stop and accelerate-go curves. An initial elevator-trim design could not trim the full CG range (26.9–38.2% MAC) at high lift coefficients; switching to a trimmable horizontal stabilizer solved it in cruise and landing.

Table 5: Performance against the RFP.

Requirement	Target	Achieved	Status
Takeoff field length	$\leq 6,000$ ft	5,234 ft	Met
Landing field length	$\leq 6,000$ ft	5,206 ft	Met
Cruise Mach (minimum)	0.85	0.85	Met
Cruise Mach (target)	0.92	$M_{crit} = 0.876$	Not met
Design range	8,000 nmi	7,849 nmi	1.9% short
Cabin accommodations	RFP list	All included	Met

Engineering takeaway. The shortfalls were traced rather than hidden: range is limited by cruise L/D (13.28 from AAA; 14 would give 8,275 nmi) and the Mach 0.92 target by the wing and tail critical Mach numbers (0.876 and 0.883), pointing to airfoil tailoring as the next iteration. A sensitivity study showed the landing distance was driven mainly by the assumed braking friction coefficient.